

Triangle Properties — Full Solutions & Explanations

Digital SAT Style | Complete Working for Every Question

Q1 — EASY

■ **Correct Answer: B) 65°**

STEP-BY-STEP SOLUTION:

- Sum of interior angles = 180° .
- Angle C = $180^\circ - 50^\circ - 65^\circ = 65^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 55° from subtraction error.
- ✗ C: 70° from wrong arithmetic.
- ✗ D: 75° confuses with a supplement.

■ **TRAP ALERT:** Students sometimes subtract from 360° instead of 180° .

■ **KEY INSIGHT:** Triangle angle sum = 180° . Always.

■ **FAST METHOD:** $180^\circ - 50^\circ - 65^\circ = 65^\circ$.

Q2 — EASY

■ **Correct Answer: B) 25**

STEP-BY-STEP SOLUTION:

- $(2x+10)+(3x-5)+(x+25) = 180$.
- $6x + 30 = 180 \rightarrow 6x = 150 \rightarrow x = 25$.
- Check: $60^\circ + 70^\circ + 50^\circ = 180^\circ$ ✓

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $x=20 \rightarrow 50^\circ + 55^\circ + 45^\circ = 150^\circ$.
- ✗ C: $x=30 \rightarrow 70^\circ + 85^\circ + 55^\circ = 210^\circ$.
- ✗ D: $x=35 \rightarrow 80^\circ + 100^\circ + 60^\circ = 240^\circ$.

■ **TRAP ALERT:** Don't forget ALL three angles must sum to 180° , not just two.

■ **KEY INSIGHT:** Set up one equation: sum of all three angle expressions = 180° .

■ **FAST METHOD:** $6x+30=180 \rightarrow x=25$.

Q3 — EASY

■ **Correct Answer: C) 75°**

STEP-BY-STEP SOLUTION:

- Exterior angle = sum of two non-adjacent interior angles.
- $130^\circ = 55^\circ + \text{other} \rightarrow \text{other} = 75^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $50^\circ = 180^\circ - 130^\circ$, confusing with the adjacent interior.
- ✗ B: 65° mixes up which angle is given.
- ✗ D: 80° is an arithmetic error.

■ **TRAP ALERT:** Computing $180^\circ - 130^\circ = 50^\circ$ gives the ADJACENT interior angle, not the non-adjacent one.

■ **KEY INSIGHT:** Exterior Angle Theorem: exterior = sum of the two remote interior angles.

■ **FAST METHOD:** $130^\circ - 55^\circ = 75^\circ$.

Q4 — EASY

■ **Correct Answer:** B) 50°

STEP-BY-STEP SOLUTION:

1. Angle A = angle B (isosceles).
2. $A + B + 80^\circ = 180^\circ \rightarrow 2A = 100^\circ \rightarrow A = 50^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $40^\circ = 80^\circ/2$, dividing the wrong angle.
- ✗ C: 60° is for equilateral triangles.
- ✗ D: 80° assumes all equal to C.

■ **TRAP ALERT:** Dividing 80° by 2 instead of dividing $(180^\circ - 80^\circ)$ by 2.

■ **KEY INSIGHT:** Isosceles: two equal angles. Sum of all three = 180° .

■ **FAST METHOD:** $(180^\circ - 80^\circ)/2 = 50^\circ$.

Q5 — EASY

■ **Correct Answer:** B) 10 cm

STEP-BY-STEP SOLUTION:

1. Pythagorean theorem: $a^2 + b^2 = c^2$.
2. $6^2 + 8^2 = 36 + 64 = 100$.
3. $c = \sqrt{100} = 10$ cm.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 7 is not a Pythagorean triple with 6 and 8.
- ✗ C: $12 = 6 + 8 - 2$, wrong method.
- ✗ D: $14 = 6 + 8$, adding legs instead of using Pythagorean theorem.

■ **TRAP ALERT:** Adding the legs ($6 + 8 = 14$) instead of using $a^2 + b^2 = c^2$.

■ **KEY INSIGHT:** Common Pythagorean triples: 3-4-5, 5-12-13, 8-15-17. 6-8-10 is $2 \times (3-4-5)$.

■ **FAST METHOD:** Recognize 6-8-10 triple ($2 \times$ the 3-4-5). ■ Desmos: $\sqrt{(36+64)}=10$.

Q6 — EASY

■ **Correct Answer:** B) 15

STEP-BY-STEP SOLUTION:

1. Similar triangles have proportional sides.
2. $DE/GH = EF/HI \rightarrow 4/10 = 6/HI$.
3. $HI = 6 \times 10/4 = 15$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $12 = 6 \times 2$, using wrong scale factor.
- ✗ C: $18 = 6 \times 3$, wrong ratio.

✗ D: $20 = 10 \times 2$, scaling the wrong side.

■ **TRAP ALERT:** Using the wrong pair of corresponding sides for the ratio.

■ **KEY INSIGHT:** In similar triangles, matching sides must correspond: $DE \leftrightarrow GH$, $EF \leftrightarrow HI$.

■ **FAST METHOD:** Scale factor = $10/4 = 2.5$. $HI = 6 \times 2.5 = 15$.

Q7 — EASY

■ **Correct Answer:** B) 55°

STEP-BY-STEP SOLUTION:

1. Sum = 180° . Angle Z = $180^\circ - 90^\circ - 35^\circ = 55^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 45° assumes equal acute angles.

✗ C: 65° is $100^\circ - 35^\circ$, wrong subtraction.

✗ D: 90° would make two right angles — impossible.

■ **TRAP ALERT:** In a right triangle, the two acute angles sum to 90° , not 180° .

■ **KEY INSIGHT:** Right triangle: acute angles sum to 90° . So $Z = 90^\circ - 35^\circ = 55^\circ$.

■ **FAST METHOD:** $90^\circ - 35^\circ = 55^\circ$.

Q8 — EASY

■ **Correct Answer:** A) Yes, because $5^2 + 12^2 = 13^2$

STEP-BY-STEP SOLUTION:

1. Check: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$.

2. Since $a^2 + b^2 = c^2$, it is a right triangle.

WHY EACH WRONG ANSWER IS WRONG:

✗ B: Adds sides instead of squaring them.

✗ C: Triangle inequality proves it's a triangle, not a RIGHT triangle.

✗ D: Odd/even is irrelevant to right triangles.

■ **TRAP ALERT:** Confusing the triangle inequality (existence) with the Pythagorean test (right angle).

■ **KEY INSIGHT:** 5-12-13 is a Pythagorean triple. Memorize the common ones.

■ **FAST METHOD:** $5^2 + 12^2 = 25 + 144 = 169 = 13^2$ ✓. Right triangle.

Q9 — MEDIUM

■ **Correct Answer:** A) 42°

STEP-BY-STEP SOLUTION:

1. $AB = BC \rightarrow$ the angles opposite these sides are equal.

2. Side AB is opposite angle C. Side BC is opposite angle A.

3. So angle C = angle A = 42° .

4. Check: angle B = $180^\circ - 42^\circ - 42^\circ = 96^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

✗ B: $69^\circ = (180^\circ - 42^\circ)/2$, treating angle A as the apex.

✗ C: 96° is angle B (the apex angle), not C.

✗ D: $138^\circ = 180^\circ - 42^\circ$, the supplement.

■ **TRAP ALERT:** Misidentifying which angles are the base angles. $AB=BC$ means the angles **OPPOSITE** these sides (C and A) are equal.

■ **KEY INSIGHT:** In isosceles triangles, equal sides → equal **OPPOSITE** angles. Identify sides carefully.

■ **FAST METHOD:** $AB=BC \rightarrow \text{angle C} = \text{angle A} = 42^\circ$.

Q10 — MEDIUM

■ **Correct Answer:** B) Right

STEP-BY-STEP SOLUTION:

1. Check: $9^2+12^2 = 81+144 = 225 = 15^2$.
2. Since $a^2+b^2 = c^2$, it is a right triangle.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: Acute needs $a^2+b^2 > c^2$; here they are equal.
- ✗ C: Obtuse needs $a^2+b^2 < c^2$; here they are equal.
- ✗ D: Equilateral needs all sides equal; $9 \neq 12 \neq 15$.

■ **TRAP ALERT:** Not checking the Pythagorean relationship and guessing based on side lengths.

■ **KEY INSIGHT:** $9\text{-}12\text{-}15 = 3 \times (3\text{-}4\text{-}5)$. Always check if sides form a Pythagorean triple.

■ **FAST METHOD:** Recognize $3 \times (3\text{-}4\text{-}5)$ triple → right triangle.

Q11 — MEDIUM

■ **Correct Answer:** A) 3:5

STEP-BY-STEP SOLUTION:

1. Area ratio = (side ratio)².
2. If area ratio = 9:25, then side ratio = $\sqrt{9:25} = 3:5$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ B: 9:25 is the area ratio, not the side ratio.
- ✗ C: 9:5 doesn't account for the square relationship.
- ✗ D: $81:625 = (9:25)^2$, which is the volume ratio for 3D figures.

■ **TRAP ALERT:** Using area ratio directly as side ratio. Must take the **SQUARE ROOT**.

■ **KEY INSIGHT:** Area ratio = k^2 . Side ratio = k . Volume ratio = k^3 .

■ **FAST METHOD:** $\sqrt{9/25} = 3/5$. Side ratio = 3:5.

Q12 — MEDIUM

■ **Correct Answer:** B) 9 cm

STEP-BY-STEP SOLUTION:

1. The midsegment is parallel to the third side and half its length.
2. Midsegment = $18/2 = 9$ cm.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $6 = 18/3$, using wrong fraction.
- ✗ C: $12 = 18 \times 2/3$, wrong formula.
- ✗ D: 18 = full side, not the midsegment.

■ **TRAP ALERT:** Forgetting the midsegment is **HALF** the third side, or confusing which side it's parallel to.

■ **KEY INSIGHT:** Midsegment Theorem: connects midpoints of two sides, is parallel to the third side, and half its length.

■ **FAST METHOD:** $18/2 = 9$ cm.

Q13 — MEDIUM

■ **Correct Answer:** C) 13

STEP-BY-STEP SOLUTION:

1. Right triangle with legs 5 and 12.
2. $AC^2 = AB^2 + BC^2 = 25 + 144 = 169$.
3. $AC = 13$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $7 = 12 - 5$, subtracting sides.
- ✗ B: 10 confuses with the 6-8-10 triple.
- ✗ D: $17 = 5 + 12$, adding sides.

■ **TRAP ALERT:** Adding legs ($5+12=17$) instead of using Pythagorean theorem.

■ **KEY INSIGHT:** 5-12-13 is a standard Pythagorean triple.

■ **FAST METHOD:** Recognize 5-12-13 triple instantly. ■ **Desmos:** $\sqrt{(25+144)} = 13$.

Q14 — MEDIUM

■ **Correct Answer:** C) 4, 5, 8

STEP-BY-STEP SOLUTION:

1. Triangle Inequality: sum of any two sides must exceed the third.
2. A) $2+3=5 < 6$. ✗
3. B) $1+4=5 < 6$. ✗
4. C) $4+5=9 > 8$, $4+8=12 > 5$, $5+8=13 > 4$. ✓
5. D) $3+3=6 < 7$. ✗

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $2+3=5 < 6$. Fails.
- ✗ B: $1+4=5 < 6$. Fails.
- ✗ D: $3+3=6 < 7$. Fails.

■ **TRAP ALERT:** Only checking one pair of sides. ALL three pairs must satisfy the inequality.

■ **KEY INSIGHT:** Triangle Inequality: $a+b > c$ for ALL combinations of sides.

■ **FAST METHOD:** Check the two SMALLEST sides: if their sum $>$ largest side, it works.

Q15 — MEDIUM

■ **Correct Answer:** B) 48 cm

STEP-BY-STEP SOLUTION:

1. Perimeter ratio = side ratio = 3:4.
2. If perimeter of ABC = 36, then perimeter of DEF = $36 \times (4/3) = 48$ cm.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $27 = 36 \times 3/4$, dividing instead of multiplying.

✗ C: $64 = 48 \times \frac{4}{3}$, applying the ratio again.

✗ D: $108 = 36 \times 3$, wrong multiplier.

■ **TRAP ALERT:** Multiplying by $\frac{3}{4}$ instead of $\frac{4}{3}$ (going from smaller to larger triangle).

■ **KEY INSIGHT:** Perimeters of similar figures are in the same ratio as corresponding sides.

■ **FAST METHOD:** $36 \times (\frac{4}{3}) = 48$ cm.

Q16 — MEDIUM

■ **Correct Answer:** B) 20

STEP-BY-STEP SOLUTION:

1. Exterior angle at R = sum of non-adjacent interior angles (P and Q).
2. $7x - 30 = (3x+10) + 2x = 5x + 10$
3. $2x = 40 \rightarrow x = 20$
4. Check: exterior = $7(20) - 30 = 110^\circ$. $P+Q = 70^\circ + 40^\circ = 110^\circ$ ✓

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $x=15 \rightarrow$ exterior $= 75^\circ$, $P+Q = 55^\circ + 30^\circ = 85^\circ$. No match.

✗ C: $x=25 \rightarrow$ exterior $= 145^\circ$, $P+Q = 85^\circ + 50^\circ = 135^\circ$. No match.

✗ D: $x=30 \rightarrow$ exterior $= 180^\circ$, impossible for a triangle.

■ **TRAP ALERT:** Using angle sum = 180° directly without recognizing the exterior angle shortcut.

■ **KEY INSIGHT:** Exterior angle = sum of two non-adjacent interior angles.

■ **FAST METHOD:** $7x - 30 = 5x + 10 \rightarrow x = 20$.

Q17 — HARD

■ **Correct Answer:** B) 24

STEP-BY-STEP SOLUTION:

1. Check if right triangle: $6^2 + 8^2 = 36 + 64 = 100 = 10^2$.
2. Yes! Right angle at A (opposite the longest side $BC=10$).
3. Area = $(\frac{1}{2})(6)(8) = 24$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $20 = (\frac{1}{2})(8)(5)$, wrong dimension.

✗ C: $30 = (\frac{1}{2})(6)(10)$, using hypotenuse as height.

✗ D: $40 = (\frac{1}{2})(8)(10)$, using hypotenuse as height.

■ **TRAP ALERT:** Using the hypotenuse (10) as a height. In a right triangle, the two LEGS are base and height.

■ **KEY INSIGHT:** First check for right triangle (Pythagorean). If right, area = $(\frac{1}{2})(\text{leg})(\text{leg})$.

■ **FAST METHOD:** 6-8-10 triple $\rightarrow$ area = $(\frac{1}{2})(6)(8) = 24$.

Q18 — HARD

■ **Correct Answer:** C) $5\sqrt{3}$

STEP-BY-STEP SOLUTION:

1. Height splits equilateral triangle into two 30-60-90 right triangles.
2. Half-base = 5. Hypotenuse = 10.

$$3. \text{ Height} = \sqrt{(10^2 - 5^2)} = \sqrt{(100 - 25)} = \sqrt{75} = 5\sqrt{3}.$$

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 5 is the half-base, not the height.
- ✗ B: $5\sqrt{2}$ is the hypotenuse of an isosceles right triangle with leg 5.
- ✗ D: $10\sqrt{3}$ confuses the formula (would need side = 20).

■ **TRAP ALERT:** Using $h = s/2$ instead of $h = (s\sqrt{3})/2$.

■ **KEY INSIGHT:** Equilateral triangle height = $(\text{side} \times \sqrt{3})/2$. This comes from the 30-60-90 triangle.

■ **FAST METHOD:** $h = (10\sqrt{3})/2 = 5\sqrt{3}$.

Q19 — HARD

■ **Correct Answer:** B) 84

STEP-BY-STEP SOLUTION:

1. $s = (13+14+15)/2 = 21$.
2. $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(21 \times 8 \times 7 \times 6)}$.
3. $= \sqrt{(21 \times 8 \times 42)} = \sqrt{(7056)} = 84$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 72 uses wrong semi-perimeter.
- ✗ C: 90 from arithmetic error.
- ✗ D: 96 confuses with a different triple.

■ **TRAP ALERT:** Forgetting to divide the perimeter by 2 for the semi-perimeter.

■ **KEY INSIGHT:** Heron's formula: $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$.

■ **FAST METHOD:** $s=21$. $21 \times 8 \times 7 \times 6 = 7056$. $\sqrt{7056} = 84$.

Q20 — HARD

■ **Correct Answer:** C) 12

STEP-BY-STEP SOLUTION:

1. Base AB lies on x-axis, length = 6.
2. Height = y-coordinate of C = 4.
3. $\text{Area} = (1/2)(6)(4) = 12$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 8 uses wrong base or height.
- ✗ B: 10 from miscounting coordinates.
- ✗ D: 16 forgets the $(1/2)$ factor.

■ **TRAP ALERT:** Forgetting the $(1/2)$ in the triangle area formula.

■ **KEY INSIGHT:** With a base on the x-axis, height = y-coordinate of the opposite vertex.

■ **FAST METHOD:** $(1/2)(\text{base})(\text{height}) = (1/2)(6)(4) = 12$. ■ **Desmos:** plot points and read dimensions.

Q21 — HARD

■ **Correct Answer:** B) 9

STEP-BY-STEP SOLUTION:

1. Median formula: $AD^2 = (2AB^2 + 2AC^2 - BC^2)/4$.

$$2. AD^2 = (2(100)+2(196)-256)/4 = (200+392-256)/4 = 336/4 = 84.$$

$$3. AD = \sqrt{84} = 2\sqrt{21} \approx 9.17 \text{ cm.}$$

4. The closest answer is B) 9.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 8 is the midpoint distance BD.

✗ C: $\sqrt{58}$ uses wrong side lengths.

✗ D: $\sqrt{65}$ confuses with another configuration.

■ **TRAP ALERT:** Confusing the median with an altitude or using the wrong formula.

■ **KEY INSIGHT:** Median formula: $m_a^2 = (2b^2+2c^2-a^2)/4$ where a is the side the median goes TO.

■ **FAST METHOD:** Plug into median formula directly.

Q22 — HARD

■ **Correct Answer:** C) 63

STEP-BY-STEP SOLUTION:

1. Longest side ratio: $27/9 = 3$.

2. All sides scale by 3: $5 \times 3 = 15$, $7 \times 3 = 21$, $9 \times 3 = 27$.

3. Perimeter = $15 + 21 + 27 = 63$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $42 = 21 \times 2$, scaling error.

✗ B: 56 scales only two sides.

✗ D: $81 = 27 \times 3$, scaling only the longest side again.

■ **TRAP ALERT:** Applying the scale factor to the perimeter directly instead of scaling each side.

■ **KEY INSIGHT:** In similar triangles, perimeter ratio = side ratio.

■ **FAST METHOD:** Perimeter of small = 21. Scale = 3. Perimeter of large = $21 \times 3 = 63$.

Q23 — HARD

■ **Correct Answer:** B) 6

STEP-BY-STEP SOLUTION:

1. In a right triangle, the altitude to the hypotenuse creates a geometric mean.

$$2. CD^2 = AD \times DB = 4 \times 9 = 36.$$

$$3. CD = 6.$$

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $5 = (4+9)/2-1$, wrong approach.

✗ C: $6.5 = (4+9)/2$, arithmetic mean not geometric.

✗ D: 7 confuses with a Pythagorean calculation.

■ **TRAP ALERT:** Using arithmetic mean $(4+9)/2=6.5$ instead of geometric mean $\sqrt{(4 \times 9)}=6$.

■ **KEY INSIGHT:** Altitude to hypotenuse: $CD^2 = AD \times DB$ (geometric mean relationship).

■ **FAST METHOD:** $CD = \sqrt{(4 \times 9)} = \sqrt{36} = 6$.

Q24 — HARD

■ **Correct Answer:** C) 6

STEP-BY-STEP SOLUTION:

1. Angle Bisector Theorem: $BD/DC = AB/AC = 12/8 = 3/2$.
2. $BD + DC = BC = 10$.
3. Let $BD = 3k$, $DC = 2k$. Then $5k = 10 \rightarrow k = 2$.
4. $BD = 3(2) = 6$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $4 = 2 \times 2$, using the wrong part.
- ✗ B: $5 = 10/2$, splitting BC equally.
- ✗ D: 7 from calculation error.

■ **TRAP ALERT:** Splitting BC equally ($BD=5$) instead of using the angle bisector ratio.

■ **KEY INSIGHT:** Angle Bisector Theorem: the bisector divides the opposite side in the ratio of the adjacent sides.

■ **FAST METHOD:** $BD/DC = 12/8 = 3/2$. $BD = 10 \times (3/5) = 6$.

Q25 — HARD

■ **Correct Answer: C) 12 ft**

STEP-BY-STEP SOLUTION:

1. Right triangle: ladder=hypotenuse=13, base=5.
2. $\text{Height}^2 = 13^2 - 5^2 = 169 - 25 = 144$.
3. Height = 12 ft.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 8 from wrong subtraction.
- ✗ B: 10 confuses with 6-8-10 triple.
- ✗ D: $14 = 13 + 1$, not Pythagorean.

■ **TRAP ALERT:** Computing $13 - 5 = 8$ instead of using the Pythagorean theorem.

■ **KEY INSIGHT:** 5-12-13 is a Pythagorean triple. Ladder problems are always right triangles.

■ **FAST METHOD:** Recognize 5-12-13 triple. Height = 12. ■ Desmos: $\sqrt{(169-25)}=12$.

TOPIC SUMMARY — Triangle Properties

Common Mistakes (Top 3)

- Using area ratio as side ratio in similar triangles. Area ratio = k^2 , side ratio = k . Always take the square root.
- Confusing which sides are opposite which angles in isosceles triangles. Equal sides $\rightarrow$ equal OPPOSITE angles.
- Adding sides instead of using Pythagorean theorem: hypotenuse $\neq$ leg + leg.

Must Remember (Top 3 Rules)

- Triangle angle sum = 180° . Exterior angle = sum of two non-adjacent interior angles.
- Pythagorean triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25. And their multiples.
- Similar triangles: side ratio = k , perimeter ratio = k , area ratio = k^2 .

Expected on DSAT

- 2–4 questions per test. Common: angle sum, similar triangles, Pythagorean theorem.
- Desmos: use $\sqrt{a^2+b^2}$ for hypotenuse calculations.